Text Generation Using Hugging Face Inference API

Develop a Python program that uses a Hugging Face model through the Inference API to generate text from user-provided prompts.

Aim

To develop a Python program that accepts user prompts, sends them to a hosted Hugging Face model using the Inference API, and displays the generated text.

Algorithm

1. Import InferenceClient and read the Hugging Face access token using hidden input.

2. Create an InferenceClient with automatic provider selection and choose the model.

3. Accept a prompt from the user inside a loop.

4. If the prompt is exit, end the program. If it is empty, request another prompt.

5. Send each valid prompt using client.chat_completion() with a user message.

6. Read response.choices[0].message.content and display the generated text.

7. Handle request failures and continue accepting prompts.

Python Program

```
# In Colab, first run: %pip install -q -U huggingface_hub
from getpass import getpass
from huggingface_hub import InferenceClient

token = getpass("Enter Hugging Face token: ").strip()
if not token:
    raise SystemExit("A token is required.")

client = InferenceClient(provider="auto", api_key=token)
model = "Qwen/Qwen2.5-7B-Instruct-1M"
print("Type 'exit' to stop.")
while True:
    prompt = input("\nEnter your prompt: ").strip()
    if prompt.lower() == "exit":
        print("Program ended.")
        break
    if not prompt:
        print("Please enter a non-empty prompt.")
        continue
    try:
        response = client.chat_completion(
            model=model,
            messages=[{"role": "user", "content": prompt}],
            max_tokens=200
        )
        text = response.choices[0].message.content
        print("Generated Output:")
        print(text or "No text returned.")
    except Exception as error:
        print("Request failed:", type(error).__name__)
        print("Check token, provider, model and quota.")
```

Input

Token: Hugging Face access token with Inference Providers permission and available quota. Model availability depends on the selected provider.

Enter Hugging Face token: [hidden input]
Prompt 1: Define AI in one sentence.
Prompt 2: List two uses of AI.
Final input: exit

Output

Sample Output

```
[ ]  # Hugging Face text generation output

Enter Hugging Face token: [hidden]
Type 'exit' to stop.

Enter your prompt: Define AI in one sentence.
Generated Output:
Artificial intelligence enables computers to perform tasks that
usually require human intelligence.

Enter your prompt: List two uses of AI.
Generated Output:
1. Detecting objects in images.
2. Recommending products to customers.

Enter your prompt: exit
Program ended.
```

Result

A Python program was developed to generate text from user-provided prompts using a hosted model through the Hugging Face Inference API.